Development of an Arduino-Based Edge AI System for Plant Health Assessment and Growth Stage Detection

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Abstract

Smallholder and backyard growers rarely have access to laboratory-grade diagnostic tools, so plant stress is often noticed only after visible damage has already reduced yield. This paper presents the concept and design of a low-cost, Arduino-based edge artificial intelligence (AI) system that assesses plant health and detects growth stage directly on-device, without depending on a constant internet connection or cloud inference server. The system combines a microcontroller with environmental sensors (soil moisture, temperature, humidity, and light intensity) and a compact image sensor, feeding a quantized convolutional neural network (CNN) that has been compressed for microcontroller deployment using TinyML techniques. The device is intended to classify leaf condition into healthy or stressed categories, estimate the plant's approximate growth stage, and issue simple, actionable alerts to the grower through a local display or mobile notification. Because the working prototype and dataset collection are still in progress, this paper reports a conceptual architecture and a methodology to be evaluated using simulated sensor and image data rather than field results. The paper reviews related literature on TinyML-based plant disease detection, growth-stage classification, and low-cost IoT agricultural monitoring, and situates the proposed system within the context of Philippine smallholder agriculture, where affordability and offline operation are practical necessities rather than optional features.

Keywords: *Arduino, Convolutional Neural Network, Edge AI, Embedded Machine Learning, Growth Stage Detection, IoT Sensors, Plant Health Monitoring, Precision Agriculture, Smallholder Farming, TinyML*

Introduction

Context and the Unsolved Problem

Plant diseases and physiological stress remain among the most persistent threats to agricultural productivity worldwide. Global estimates attribute a substantial share of annual crop losses to disease and pest pressure, with figures in the range of 20–40% depending on the crop and region (Savary et al., as cited in "Affordable Precision Agriculture," n.d.). The impact of these losses falls hardest on smallholder farmers and backyard growers, who typically lack access to trained plant pathologists, laboratory diagnostics, or the capital needed for large-scale precision agriculture equipment. In such settings, a disease or nutrient deficiency may go unnoticed until visible symptoms have already spread through a planting bed, by which point the window for effective, low-cost intervention has usually closed.

Backyard growers, urban gardeners, and smallholder farmers in the Philippines and similar contexts frequently lack affordable, offline tools to (a) detect early signs of plant stress or disease and (b) track a plant's growth stage in order to time interventions such as fertilization, pruning, or harvest. Existing precision-agriculture solutions are either too expensive, require continuous internet connectivity and cloud subscriptions, or focus narrowly on disease classification without addressing growth-stage

tracking. This creates a need for a single, low-cost, offline-capable device that can perform both functions using on-device (edge) inference.

Why the Problem Is Important

Early detection matters because the cost of intervention rises as stress spreads: a problem caught at the first signs can often be corrected cheaply, while one noticed late may already have reduced yield across a whole bed. Growth-stage tracking matters for the same reason, since fertilization, pruning, and harvest are most effective when timed to the plant's actual stage rather than guessed. For Philippine smallholders, the constraints are practical rather than theoretical: many work where connectivity is unreliable and recurring data or subscription costs are hard to justify. A tool that is cheap, works offline, and gives simple, actionable alerts could therefore make early, well-timed intervention available to growers who are currently excluded from precision agriculture.

Related Work

Image-Based Plant Disease Detection

Advances in machine learning, particularly convolutional neural networks (CNNs) trained on large, labeled image datasets such as PlantVillage, have made it possible to classify plant diseases from leaf images with high accuracy (Mohanty et al., 2016; Hughes & Salathé, 2015). Later work has continued to examine the architectural trade-offs involved in crop disease classification ("AgriPath," n.d.). However, most of these models were originally designed for cloud or smartphone-based inference, which assumes a stable internet connection, sufficient device processing power, and, often, a recurring data cost that is not realistic for many rural or resource-constrained users.

TinyML on Microcontrollers

The emergence of TinyML, the practice of compressing and running machine learning models directly on microcontrollers, has begun to close this gap (Warden & Situnayake, 2019). Frameworks such as TensorFlow Lite Micro (David et al., 2021) and platforms such as Edge Impulse now allow CNNs to be quantized and deployed on inexpensive boards such as the Arduino Nano 33 BLE Sense or ESP32-CAM, enabling fully offline, battery-friendly inference at the point of need. Hobbyist projects have likewise applied TinyML to plant health on Arduino-class hardware (Rust, n.d.).

Recent studies show this approach is technically viable for agriculture. Khan and colleagues deployed a MobileNetV3-Small model for tomato leaf disease detection that, after 8-bit quantization, achieved 99.5% test accuracy while shrinking to 3.8 MB, small enough to run on a range of edge platforms ("Automated Tomato Leaf Disease Detection," n.d.). A maize-focused study deployed a custom CNN on an Arduino Nicla Sense ME and reported an inference latency of only 7.60 ms using 726.6 KB of RAM and 344.7 KB of flash ("TinyML for Smart Agriculture," 2024). Similar results have been reported for apple and mango leaf disease detection using MobileNetV2 on TinyML hardware, with accuracy above 93% (Attri et al., 2025).

Growth-Stage Classification

Growth-stage classification has been explored as a distinct but related task. Deep learning methods for image-based plant growth monitoring have been reviewed broadly ("Deep Learning for Image-Based

Plant Growth Monitoring," 2022), and lightweight CNN backbones have been applied to crops such as strawberries (Li et al., 2024) and soybeans ("DELTA-SoyStage," n.d.). Classical feature-extraction methods such as Gabor filters, combined with transfer learning, have also been used for real-time, low-resource growth-stage estimation ("Feature Extraction for Plant Growth Estimation," n.d.).

Low-Cost IoT Agricultural Monitoring

On the sensing side, low-cost systems have been built to monitor soil water status with proximal sensors and LoRa communication for irrigation (Dafonte et al., 2024), to track soil water content and multiple soil-air-vegetation parameters on open hardware ("A Novel Low-Cost Open-Hardware Platform," n.d.), and to monitor soil and crops with low-cost smart sensors ("IoT-Enabled Soil and Crop Monitoring System," 2025). In the Philippine context, researchers have examined the extent and inclusiveness of digital technology in agri-food systems (Philippine Institute for Development Studies, 2023) and the open innovation capability of smallholder farmers (Jalotjot & Tokuda, 2024), while industry sources describe a growing range of smart farming innovations in the country (Farmonaut, 2025).

Why the Problem Has Not Been Solved

Despite this progress, several gaps remain. First, most high-accuracy models still assume cloud or smartphone inference, which excludes growers without reliable connectivity. Second, disease or stress detection and growth-stage estimation have largely been studied as separate tasks, each with its own model and hardware. Third, models trained on PlantVillage-style images, captured under controlled conditions, often perform worse on field images with real lighting and backgrounds. Fourth, sensor-based IoT systems measure the environment but cannot see the leaf itself. Combining stress/health classification with growth-stage estimation and environmental sensing on a single embedded device has, to the researchers' knowledge, not yet been widely demonstrated at the hobbyist/smallholder price point that this study targets.

Proposed Work

Objectives

General objective. To design and conceptually validate an Arduino-based edge AI system capable of assessing plant health and detecting plant growth stage using on-device machine learning inference, without reliance on continuous cloud connectivity.

Specific objectives. This study specifically aims to:

1. Design a low-cost hardware architecture combining a microcontroller, environmental sensors, and a compact image sensor suitable for on-device plant monitoring.
2. Develop and describe a lightweight, quantized CNN pipeline capable of running within the memory and processing constraints of a microcontroller-class device.
3. Define a growth-stage classification scheme appropriate for common backyard/smallholder crops (e.g., seedling, vegetative, flowering, fruiting/maturity).
4. Propose a rule-based or threshold-based alert mechanism that fuses environmental sensor readings (soil moisture, temperature, humidity, light) with image-based classification output.

5. Evaluate the proposed system's feasibility using simulated sensor and image data, given that full field deployment and dataset collection are beyond the current scope of this concept paper.

Methods

This concept paper follows a design-and-simulate methodology, appropriate to its current stage of development ahead of full hardware fabrication and field data collection.

Design phase. The hardware architecture and bill of materials were selected based on component choices validated in the reviewed literature (e.g., Arduino Nano 33 BLE Sense-class boards, capacitive soil sensors, DHT-series temperature/humidity sensors), prioritizing components with an existing track record in low-cost agricultural monitoring projects.

Model development plan. A MobileNet-family CNN will be selected as the base architecture given its consistent performance in the reviewed TinyML disease-detection studies. The model will be pre-trained via transfer learning on a public dataset (PlantVillage) and fine-tuned on a smaller, purpose-collected set of images for the target crops, addressing the lab-vs-field domain gap noted in the related work. The trained model will then undergo post-training 8-bit quantization and conversion to a TensorFlow Lite Micro-compatible format for on-device deployment.

Simulated evaluation. Because a physical prototype and field-collected dataset are not yet available, the proposed pipeline will be evaluated using simulated data: synthetic sensor readings generated within realistic ranges for soil moisture, temperature, humidity, and light, paired with a held-out subset of publicly available leaf images. These will be used to approximate expected classification behavior and to estimate model size, memory footprint, and inference latency on the target microcontroller, benchmarked against the figures reported in comparable published deployments.

Evaluation criteria. The simulated pipeline will be assessed against four criteria: (a) classification accuracy on the validation split, (b) compressed model size relative to the target board's flash capacity, (c) estimated inference latency, and (d) the false-alert rate of the sensor-fusion logic under simulated edge-case environmental conditions (e.g., simulated drought stress vs. simulated early disease onset).

Preliminary Results

No measured results are available yet, because the physical prototype and local dataset have not been built. As a feasibility baseline, the design targets below are drawn from comparable published deployments. They are reference figures from other studies, not results of this project.

Measure	Reported in comparable work	Design target for this system
Inference latency	7.60 ms (maize CNN, Arduino Nicla Sense ME)	Under 10 ms per image
Flash footprint	344.7 KB (maize); 3.8 MB (tomato, MobileNetV3-Small)	Under 1 MB, to fit Nano 33 BLE Sense-class boards
RAM usage	726.6 KB (maize)	Within the target board's available RAM

Measure	Reported in comparable work	Design target for this system
Classification accuracy	99.5% (tomato); above 93% (apple/mango)	Above 90% on the validation split

The results of the simulated evaluation, covering all four evaluation criteria, will be added to this section once completed.

Anticipated Challenges and Mitigations

Challenge	Mitigation
Lab-vs-field domain gap: models trained on PlantVillage-style images may perform poorly on real field photos.	Fine-tune on locally collected images taken under real lighting and backgrounds; apply data augmentation (brightness, rotation, background variation).
Tight memory limits: microcontroller boards have very little flash and RAM.	Use a small MobileNet variant with low-resolution input, apply 8-bit quantization, and check model size against the board's limits before deployment.
Two tasks on one board: health classification and growth-stage detection both need inference.	Use one shared backbone with two output heads, or run two small models in turn, since readings are only needed periodically.
False alerts: drought stress and early disease can look similar.	Raise an alert only when image and sensor evidence agree, require repeated readings before alerting, and tune thresholds on the simulated edge cases.
Variable lighting: poor light lowers image quality.	Capture images at fixed times and use the light sensor reading to skip images taken in poor light.
Small local dataset: collecting and labeling images takes time.	Rely on transfer learning and write a clear labeling guide for growth stages so that every team member labels consistently.
Power use in the field: continuous sensing drains batteries.	Put the board to sleep between readings and wake it only at set intervals.

Concrete Deliverables and Expected Completion Times

Phase	Deliverable	Expected completion
1	Concept paper (this document)	Week 0 (current)
2	Simulated evaluation results for all four criteria	Weeks 1–3

Phase	Deliverable	Expected completion
3	Assembled hardware prototype with all sensors tested	Weeks 3–5
4	Local image dataset collected and labeled for health and growth stage	Weeks 4–8
5	Fine-tuned, quantized model deployed on the board	Weeks 8–10
6	Bench testing report (accuracy, latency, false-alert rate on the device)	Weeks 10–12
7	Final paper with measured results, ahead of any field trial	Weeks 13–14

References

1. Attri, I., Awasthi, L. K., & Sharma, T. P. (2025). TinyML for plant disease detection: Efficient edge AI solutions for apple and mango leaves. *Procedia Computer Science, 258*, 2870–2877. https://doi.org/10.1016/j.procs.2025.04.547
2. David, R., Duke, J., Jain, A., Janapa Reddi, V., Jeffries, N., Li, J., Kreeger, N., Nappier, I., Natraj, M., Regev, S., Rhodes, R., Wang, T., & Warden, P. (2021). TensorFlow Lite Micro: Embedded machine learning on TinyML systems. *Proceedings of Machine Learning and Systems (MLSys), 3*, 800–811. https://proceedings.mlsys.org/paper_files/paper/2021/hash/6c44dc73014d66ba49b28d483a8f8b0d-Abstract.html
3. Dafonte, J., González, M. Á., Comesaña, E., Teijeiro, M. T., & Cancela, J. J. (2024). Soil water status monitoring system with proximal low-cost sensors and LoRa technology for smart water irrigation in woody crops. *Sensors, 24*(24), 8104. https://doi.org/10.3390/s24248104
4. Farmonaut. (2025, August 13). *Smart farming in the Philippines: 2025 innovations.* https://farmonaut.com/asia/smart-farming-in-the-philippines-2025-innovations
5. Hughes, D. P., & Salathé, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics through machine learning and crowdsourcing. *arXiv preprint arXiv:1511.08060.* https://arxiv.org/abs/1511.08060
6. Jalotjot, H. C., & Tokuda, H. (2024). Exploring smallholder farmers' open innovation capability: A structural equation modeling approach. *ScienceDirect.* https://www.sciencedirect.com/science/article/pii/S2199853124000994
7. Li, Z., Wang, J., Gao, G., Lei, Y., Zhao, C., Wang, Y., Bai, H., Liu, Y., Guo, X., & Li, Q. (2024). SGSNet: A lightweight deep learning model for strawberry growth stage detection. *Frontiers in Plant Science.* https://doi.org/10.3389/fpls.2024.1491706
8. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science, 7*, Article 1419. https://doi.org/10.3389/fpls.2016.01419

9. Philippine Institute for Development Studies. (2023). *Transforming Philippine agri-food systems with digital technology: Extent, prospects, and inclusiveness* [Discussion paper]. https://pids.gov.ph/publication/discussion-papers/transforming-philippine-agri-food-systems-with-digital-technology-extent-prospects-and-inclusiveness
10. Rust, E. (n.d.). *Determining a plant's health with TinyML.* Hackster.io. https://www.hackster.io/gatoninja236/determining-a-plant-s-health-with-tinyml-085003
11. Affordable precision agriculture: A deployment-oriented review of low-cost, low-power edge AI and TinyML for resource-constrained farming systems. (n.d.). *arXiv.* https://arxiv.org/pdf/2603.15085
12. Automated tomato leaf disease detection and alert system using Internet of Things and TinyML. (n.d.). *Discover Internet of Things.* https://link.springer.com/article/10.1007/s43926-025-00257-8
13. Deep learning for image-based plant growth monitoring: A review. (2022). *International Journal of Engineering and Technology Innovation, 12*(3), 225–246. https://www.researchgate.net/publication/361567510
14. A novel low-cost open-hardware platform for monitoring soil water content and multiple soil-air-vegetation parameters. (n.d.). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4239887/
15. TinyML for smart agriculture: Comparative analysis of TinyML platforms and practical deployment for maize leaf disease identification. (2024). https://www.sciencedirect.com/science/article/pii/S2772375524000959
16. IoT-enabled soil and crop monitoring system using low-cost smart sensors for precision agriculture. (2025). https://doi.org/10.3390/ECSA-12-26537
17. DELTA-SoyStage: A lightweight detection architecture for full-cycle soybean growth stage monitoring. (n.d.). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12694477/
18. Feature extraction for plant growth estimation. (n.d.). *arXiv.* https://arxiv.org/html/2606.11966
19. AgriPath: A systematic exploration of architectural trade-offs for crop disease classification. (n.d.). *arXiv.* https://arxiv.org/pdf/2603.13354
20. Warden, P., & Situnayake, D. (2019). *TinyML: Machine learning with TensorFlow Lite on Arduino and ultra-low-power microcontrollers.* O'Reilly Media.